

**MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL**

Paper Code : PECAIS01A/PECAIDS01B/PECAIML 501A Cloud Computing

UPID : 005932

Time Allotted : 3 Hours

Full Marks : 70

*The Figures in the margin indicate full marks.**Candidate are required to give their answers in their own words as far as practicable***Group-A (Very Short Answer Type Question)**

1. Answer any ten of the following :

[1 x 10 = 10]

(i) Choose the correct option.

1. Azure's architecture, an application can run locally, run in the cloud or some combination of both.
2. Azure Windows Services Platform API uses the industry standard REST, HTTP and XML protocols that are part of SOA cloud infrastructure.
3. Azure service platform hosts runtime versions of .NET Framework applications written in any of the languages like C++, Java.
4. All of the above.

(ii) Although encryption protects your data from unauthorized access, it does nothing to prevent data loss.

1. True
2. False
3. Not sure
4. None of the above

(iii) The most commonly used message-passing format is an Extensible Markup Language (XML) document using SOAP

1. true
2. false
3. not sure
4. none of the above

(iv) Network management systems are often described in terms of the acronym FCAPS. FCAPS stands for which of the following?

1. Fault, Configuration, Accounting, Performance, Security.
2. Feature, Cloud, Access, Preference, Service
3. Fault, Cloud, Accounting, Performance, Security
4. Feature, Configuration, Access, Preference, Service

(v) The advantage of Open SaaS over SaaS is

1. More security
2. Fast Access
3. Cheap deployment
4. None of the above

(vi) Google App Engine is a _____ cloud-based Web hosting service on Google's infrastructure. Fill in the blank with appropriate option from below:

1. SaaS
2. PaaS
3. IaaS
4. CaaS

(vii) According to NIST model definition there are _____ cloud deployment models.

1. 4
2. 3
3. 6
4. 7

(viii) Hybrid applications where part of the application is on a local system and part is on the cloud are known as

1. Cloud connection
2. Cloud deployment
3. Cloud bursting
4. Cloud transportation

(ix) The core functions of IDaaS are

1. a data store, a query engine, a policy engine that maintains data integrity
2. a data store, PaaS, XML file
3. SAML, SOA, Datastore
4. None of the above.

(x) A cloud computing infrastructure where a development environment is created on which applications can be built

1. IaaS
2. PaaS
3. SaaS
4. IDaaS

(xi) Which of the following are true about virtualization?

1. Assigns a logical name for a physical resource and then provides a pointer to that physical resource.
2. Provides a means to manage resources efficiently.
3. Is dynamic in that mapping can be assigned based on rapidly changing conditions.
4. All of the above.

(xii) Google's search technology is based on automated page indexing and information retrieval by Web crawlers, also called <https://www.makaut.com>

1. Worms
2. Spiders
3. Links
4. Cases

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. What do you mean by virtualization? Discuss some virtualization technologies. [5]
3. Describe the working of some Amazon Web Service Components and Services like Amazon EC2 and Amazon S3. [5]
4. What are the differences between community and hybrid models? which one is more preferable? [5]
5. Describe IaaS with an example. How does it work? [5]
6. What do you mean by workload? Discuss key features of Amazon Elastic Compute Cloud (EC2). [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. (a) What are the security boundaries in a CSA Cloud Reference Model? [5]
(b) Describe the CSA Cloud Reference Model with security boundaries with a diagram. [10]
8. (a) What are the unique characteristics of an ideal cloud computing model? [5]
(b) How is cost of a cloud computing deployment model calculated or estimated? [5]

- (c) What are the parameters specified in service level agreements(SLAs)? [5]
9. (a) What is abstraction and why is it required? [5]
- (b) Differentiate between Full Virtualization and Para Virtualization. [5]
- (c) Define Hypervisor in cloud computing and their types. [5]
10. (a) Explain the Brokered cloud storage access mechanism. [5]
- (b) What are the features of cloud storage backup solutions? [5]
- (c) Describe the security boundary. [5]
11. (a) What is composability? [5]
- (b) What do you mean by scalability and multitenancy? [5]
- (c) Discuss Identity as a Service (IDaaS) . [5]

*** END OF PAPER ***